A Fixed-Index Normalization for Integer Programs with Bounded Full-Size Subdeterminants

Draft note

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Abstract

This note records partial progress on the fixed-Δ bounded-subdeterminant integer programming problem appearing in the work of Nägele, Santiago, and Zenklusen. The note does *not* give a polynomial-time algorithm for the full problem. Its contribution is a rigorous polynomial-time normalization: every instance with full column rank and all full-size row subdeterminants bounded by a fixed constant Δ can be transformed exactly into an integer program over a sublattice of $\mathbb{Z}^n$ of index at most Δ, with a rational constraint matrix that contains an identity row basis and whose minors of all orders are bounded by Δ. The lattice condition can further be represented by congruences involving at most $\lfloor \log_2 \Delta \rfloor$ distinguished coordinates. Consequently, an algorithm for the resulting rational all-minor-bounded fixed-congruence class would solve the original open problem. This last algorithmic ingredient remains open here.

1 Status and problem statement

The result below is a reduction and normalization theorem. It is not a solution of the open problem. More precisely, it proves that the original integer program can be rewritten in a form in which the non-unimodular information is confined to a finite quotient of size at most Δ, and even to at most $\lfloor \log_2 \Delta \rfloor$ residue coordinates. The remaining task–solving the normalized rational fixed-congruence problem in polynomial time–is not accomplished in this note.

For a matrix $A \in \mathbb{Z}^{m\times n}$ of full column rank, write

$$\delta_n(A) := \max\{|\det A_I| : I \subseteq [m],\ |I| = n\},$$

where A_I denotes the $n \times n$ matrix formed by the rows indexed by I. The fixed-Δ problem is the following.

Problem (IP_Δ)**.** For each fixed positive integer Δ, decide whether there is a polynomial-time algorithm for

$$\min\{c^\top x : Ax \le b,\ x \in \mathbb{Z}^n\},$$

where $A \in \mathbb{Z}^{m\times n}$ has full column rank and $\delta_n(A) \le \Delta$.

We assume throughout that $b \in \mathbb{Z}^m$ and $c \in \mathbb{Z}^n$, although allowing rational objective coefficients would not change the reduction. Feasibility, infeasibility, unboundedness, and optimal solutions are all preserved by the transformations below.

2 Basis-coordinate normalization

The first step is to choose any nonsingular row basis of A. This converts the full-size-minor bound into an all-minor bound after a rational change of coordinates.

Theorem 1 (Basis-coordinate normalization). *Let $A \in \mathbb{Z}^{m\times n}$ have full column rank and let $\delta_n(A) \le \Delta$. Let B be any nonsingular $n \times n$ row submatrix of A, and put*

$$d := |\det B|, \qquad L := B\mathbb{Z}^n \subseteq \mathbb{Z}^n, \qquad \Lambda := AB^{-1}, \qquad \gamma := B^{-\top}c.$$

Then the original integer program is exactly equivalent, via $y = Bx$, to

$$\min\{\gamma^\top y : \Lambda y \le b,\ y \in L\}.$$

Moreover:

(a) $[\mathbb{Z}^n : L] = d \le \Delta$;

(b) Λ *contains I_n as a row submatrix;*

(c) *every entry of Λ and γ has denominator dividing d;*

(d) *every square minor of Λ, of every order, has absolute value at most Δ/d, and hence at most Δ.*

The construction is computable in polynomial time, with rational encoding length polynomial in the input size.

Proof. Since B is an integer nonsingular row submatrix of A, the determinant $d = |\det B|$ is a positive integer. The assumption $\delta_n(A) \le \Delta$ gives $d \le \Delta$.

Set $y = Bx$. Then $x \in \mathbb{Z}^n$ if and only if $y \in B\mathbb{Z}^n = L$, and in that case $x = B^{-1}y$. The inequalities and objective transform as

$$Ax \le b \iff AB^{-1}y \le b \iff \Lambda y \le b,$$

whereas

$$c^\top x = c^\top B^{-1}y = (B^{-\top}c)^\top y = \gamma^\top y.$$

Thus the feasible sets and objective values correspond bijectively.

The lattice index is

$$[\mathbb{Z}^n : B\mathbb{Z}^n] = |\det B| = d.$$

The rows of Λ corresponding to the chosen row basis B are the rows of $BB^{-1} = I_n$, so Λ contains I_n as a row submatrix. Since

$$B^{-1} = \frac{\operatorname{adj}(B)}{\det B},$$

all entries of B^{-1} have denominator dividing d, and therefore the same is true for $\Lambda = AB^{-1}$ and $\gamma = B^{-\top}c$.

It remains to prove the minor bound. Let $M = \Lambda_{R,S}$ be a $t \times t$ minor of Λ, where $R \subseteq [m]$, $S \subseteq [n]$, and $|R| = |S| = t$. The cases $t = 0$ and $t = n$ are immediate from the argument below, so assume $1 \le t \le n$.

Label the identity rows of Λ by $1, \dots, n$, so that identity row j is $e_j^\top$. If the selected rows R already contain an identity row $e_j^\top$ with $j \notin S$, then the corresponding row of M is zero and $\det M = 0$. Otherwise, adjoin to the selected rows R all identity rows $e_j^\top$ with $j \notin S$. This gives n

distinct row indices of Λ. If the columns are ordered with S first and $[n] \setminus S$ second, the resulting $n \times n$ submatrix C has block form

$$C = \begin{pmatrix} M & * \\ 0 & I_{n-t} \end{pmatrix},$$

up to row and column permutations. Hence $|\det C| = |\det M|$.

The matrix C is an $n \times n$ row submatrix of $\Lambda = AB^{-1}$. Thus $C = A_I B^{-1}$ for some n-element row set $I \subseteq [m]$. Therefore

$$\det C = \frac{\det A_I}{\det B},$$

and so

$$|\det M| = |\det C| = \frac{|\det A_I|}{d} \leq \frac{\delta_n(A)}{d} \leq \frac{\Delta}{d}.$$

This proves the claimed all-minor bound. □

Remark 1 (Maximum-determinant basis)*.* If one chooses B with $|\det B| = \delta_n(A)$, then Theorem 1 gives

$$|\det M| \leq 1$$

for every square minor M of $\Lambda = AB^{-1}$. This is a useful structural sharpening but not, by itself, an algorithmic step in the reduction above: the theorem only needs an arbitrary row basis, which is easy to find by Gaussian elimination.

3 Compressing the lattice condition to few coordinates

The lattice L in Theorem 1 has index at most Δ. A finite abelian group of order d can be generated by at most $\lfloor \log_2 d \rfloor$ elements. Applying this to $\mathbb{Z}^n/L$ shows that only constantly many coordinate residues are needed to express the lattice constraint.

Lemma 1 (Small coordinate generating set for the quotient)**.** *Let $L \subseteq \mathbb{Z}^n$ be a full-rank sublattice of index d. Then there is a subset $J \subseteq [n]$ with*

$$|J| \leq \lfloor \log_2 d \rfloor$$

whose coordinate classes generate the quotient group $G := \mathbb{Z}^n/L$. More explicitly, if $E = [n] \setminus J$, then there are an integer matrix $R \in \mathbb{Z}^{J \times E}$ and a full-rank lattice $K \subseteq \mathbb{Z}^J$ of index d such that

$$y \in L \iff y_J + Ry_E \in K.$$

Writing $K = Q\mathbb{Z}^J$ for an integer nonsingular matrix $Q \in \mathbb{Z}^{J \times J}$, this becomes

$$y \in L \iff y_J + Ry_E \in Q\mathbb{Z}^J, \qquad |\det Q| = d.$$

The data J, R, Q can be computed in polynomial time from any integer basis of L.

Proof. Let $\pi : \mathbb{Z}^n \to G := \mathbb{Z}^n/L$ be the quotient map. The classes $\pi(e_1), \ldots, \pi(e_n)$ generate G. Starting from the trivial subgroup, greedily add a coordinate class not contained in the subgroup generated so far. Each addition increases the subgroup order by a factor of at least 2. Since $|G| = d$, this process stops after at most $\lfloor \log_2 d \rfloor$ additions. Let J be the chosen set of coordinates.

Because the classes $\{\pi(e_j) : j \in J\}$ generate G, for every $i \in E := [n] \setminus J$ there is a vector $r_i \in \mathbb{Z}^J$ such that

$$\pi(e_i) = \pi\left(\sum_{j \in J} (r_i)_j e_j\right).$$

Let R be the matrix with columns r_i. Define

$$K := \left\{ u \in \mathbb{Z}^J : \pi\left(\sum_{j \in J} u_j e_j\right) = 0 \right\}.$$

Equivalently, K is the kernel of the surjective homomorphism $\mathbb{Z}^J \to G$ induced by the selected coordinate classes. Hence K is a full-rank sublattice of $\mathbb{Z}^J$ and

$$[\mathbb{Z}^J : K] = |G| = d.$$

For any $y \in \mathbb{Z}^n$,

$$\pi(y) = \pi\left(\sum_{j \in J} y_j e_j + \sum_{i \in E} y_i e_i\right) = \pi\left(\sum_{j \in J} (y_J + R y_E)_j e_j\right).$$

Thus $y \in L$ if and only if $y_J + R y_E \in K$.

Finally, any full-rank sublattice $K \subseteq \mathbb{Z}^J$ has an integer basis matrix Q with $K = Q\mathbb{Z}^J$ and $|\det Q| = [\mathbb{Z}^J : K] = d$. Standard Hermite or Smith normal form algorithms compute the greedy subgroup indices, the representing vectors r_i, and a basis Q in polynomial time. $\square$

Corollary 1 (Fixed-congruence normalized form)**.** *For fixed Δ, every instance of* (IP$_\Delta$) *reduces in polynomial time to the following exact form:*

$$\min\{\gamma^\top y : \Lambda y \le b,\ y \in \mathbb{Z}^n,\ y_J + R y_E \in Q\mathbb{Z}^J\}, \tag{FC$_\Delta$}$$

where $E = [n] \setminus J$ and:

(i) $|J| \le \lfloor \log_2 \Delta \rfloor$;

(ii) *Q is a nonsingular integer $|J| \times |J|$ matrix with $|\det Q| \le \Delta$;*

(iii) *Λ contains I_n as a row submatrix;*

(iv) *every square minor of Λ has absolute value at most Δ;*

(v) *every denominator appearing in Λ and γ divides an integer $d \le \Delta$.*

The map from solutions of (FC$_\Delta$) *back to solutions of the original problem is $x = B^{-1}y$.*

Proof. Apply Theorem 1 to obtain Λ, γ, L and then apply Lemma 1 to the lattice L. $\square$

4 What this proves, and what remains open

Corollary 1 is the main unconditional result of this note. It is a structural reduction, not a complete algorithm for fixed $\Delta > 1$. The missing algorithmic statement is the following residual problem.

Conjecture 1 (Residual rational fixed-congruence problem)**.** *For each fixed positive integer Δ, there is a polynomial-time algorithm for every instance of* (FC_Δ) *satisfying conditions* (i)–(v) *in Corollary 1.*

Proposition 1 (Conditional completion)**.** *If Conjecture 1 holds, then* (IP_Δ) *is polynomial-time solvable for every fixed Δ.*

Proof. Given an instance of (IP_Δ), compute the normalized instance in Corollary 1. Run the assumed algorithm for Conjecture 1. If it returns an optimal normalized solution y^*, return $x^* = B^{-1}y^*$. The bijection in Theorem 1 preserves feasibility and objective values, so x^* is optimal for the original instance. Infeasibility and unboundedness are preserved for the same reason. □

The proposition is intentionally conditional. It should not be read as a solution of the open problem, because Conjecture 1 is essentially the remaining hard part. The rationality of Λ and the fixed-index lattice congruence cannot be discarded without further argument.

5 Two cautions

5.1 The case $\Delta = 1$ is normalized TU, not necessarily raw TU

When $\Delta = 1$, Theorem 1 gives $d = 1$, so $L = \mathbb{Z}^n$, and $\Lambda = AB^{-1}$ is an integral matrix all of whose minors have absolute value at most 1. Hence Λ is totally unimodular, and the normalized problem is solvable by linear programming.

This does not mean that the original matrix A is necessarily totally unimodular. For example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

has $\delta_2(A) = 1$, but has a 1×1 minor equal to 2. Thus the correct statement is that an appropriate basis-coordinate normalization is TU when $\Delta = 1$.

5.2 The nonexceptional raw block need not be TU

A tempting proof strategy is to use the few exceptional congruence coordinates from Lemma 1 and then hope that the remaining raw coefficient block is totally unimodular. This is false.

Example 1. Let

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 2 & 3 \end{pmatrix}.$$

The absolute values of the 2×2 row determinants are $3, 3, 0$, so $\delta_2(A) = 3$. The one-column block obtained by retaining only the first coordinate contains an entry 2, and therefore cannot be totally unimodular. The normalization theorem applies to $\Lambda = AB^{-1}$, not to arbitrary raw subblocks of A.

6 Conclusion

The note obtains genuine but limited partial progress. It proves the exact reduction

$$\text{bounded full-size subdeterminants} \\ \Longrightarrow \\ \text{rational all-minor-bounded system plus a fixed-index congruence,}$$

with at most $\lfloor \log_2 \Delta \rfloor$ congruence coordinates. It does not prove that the normalized class is polynomial-time solvable. Therefore it does not solve the fixed-Δ bounded-subdeterminant integer programming problem. Any proof of Conjecture 1, or any other polynomial-time algorithm for the normalized class of Corollary 1, would complete the solution.

References

[1] M. Nägele, R. Santiago, and R. Zenklusen. Congruency-constrained TU problems beyond the bimodular case. *Mathematics of Operations Research* 49(3):1303–1348, 2024. doi:10.1287/moor.2023.1381.

[2] P. Nuti et al. Open Problems in Operations Research, problem W4386289171_p0. Problem page.